Example 5.16 Find the sum up to n terms of the series: $1 + \frac{6}{7} + \frac{11}{49} + \frac{16}{343} + \cdots$.

Solution:
Here $a = 1$, $d = 5$ and $r = \frac{1}{7}$.

$$\begin{aligned} S_n &= \frac{a-(a+(n-1)d)r^n}{1-r} + dr\left(\frac{1-r^{n-1}}{(1-r)^2}\right) \\ &= \frac{1-(1+5(n-1))(\frac{1}{7})^n}{1-\frac{1}{7}} + 5 \times \frac{1}{7}\left(\frac{(1-\left(\frac{1}{7}\right)^{n-1})}{(1-\frac{1}{7})^2}\right) \\ &= \frac{1-\frac{5n-4}{7^n}}{\frac{6}{7}} + \frac{\frac{5}{7}(7^{n-1}-1)}{7^{n-1}(\frac{6}{7})^2} \\ &= \frac{7^n-5n+4}{7^{n-1}6} + \frac{5(7^{n-1}-1)}{7^{n-2}36} \end{aligned}$$

5.5.2 Telescopic Summation for Finite Series

Telescopic summation is a more general method used for summing a series either for finite or infinite terms. This technique expresses sum of n terms of a given series just in two terms, usually first and last term, by making the intermediate terms cancel each other. After canceling intermediate terms, we bring the last term which is far away from the first term very close to the first term. So this process is called "Telescopic Summation".

Example 5.17 Find the sum of the first n terms of the series $\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \ldots$

Solution:
Let t_k denote the kth term of the given series. Then $t_k = \frac{1}{\sqrt{k}+\sqrt{k+1}}$. If we are successful in writing the kth term as a difference of two expressions, then we can solve using this technique. We have

$$t_k = \frac{1}{\sqrt{k}+\sqrt{k+1}} = \frac{\sqrt{k}-\sqrt{k+1}}{(\sqrt{k}+\sqrt{k+1})(\sqrt{k}-\sqrt{k+1})} = \frac{\sqrt{k}-\sqrt{k+1}}{k-(k+1)} = \sqrt{k+1}-\sqrt{k}$$

Thus

$$t_1 + t_2 + \cdots + t_n = (\sqrt{2}-\sqrt{1}) + (\sqrt{3}-\sqrt{2}) + \cdots + (\sqrt{n+1}-\sqrt{n}) = \sqrt{n+1}-1$$

Example 5.18 Find $\sum_{k=1}^{n} \frac{1}{k(k+1)}$.

Solution:
Let t_k denote the kth term of the given series. Then $t_k = \frac{1}{k(k+1)}$. By using partial fraction we get

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Thus

$$t_1 + t_2 + \cdots + t_n = \left(1-\frac{1}{2}\right) + \left(\frac{1}{2}-\frac{1}{3}\right) + \left(\frac{1}{3}-\frac{1}{4}\right) + \cdots + \left(\frac{1}{n}-\frac{1}{n+1}\right) = 1 - \frac{1}{n+1}.$$